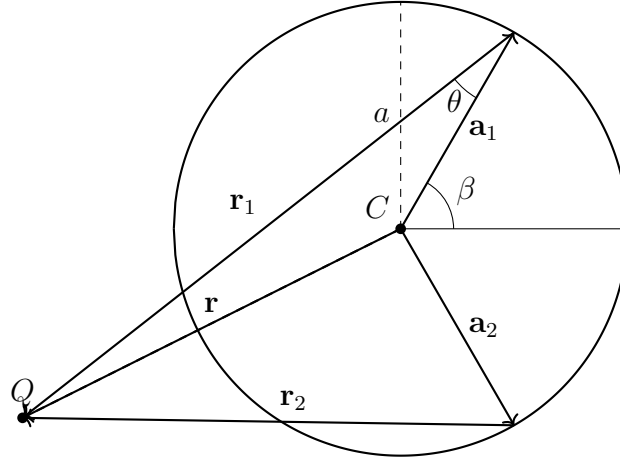




A força $d\mathbf{F}_1$ é então:

$$d\mathbf{F}_1 = k \frac{Qdq}{r'^3} \mathbf{r}_1$$

Escrevendo $\mathbf{r}_1$ em termos de $\mathbf{i}$, $\mathbf{j}$ e $\mathbf{k}$:

$$\mathbf{r}_1 = \mathbf{r} - \mathbf{a}_1$$

$$\mathbf{r}_1 = -a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k}$$

Definindo uma distribuição de carga uniforme λ na parte superior:

$$\lambda = \frac{dq}{d\beta a} \Rightarrow dq = \lambda d\beta a$$

Ou seja:

$$d\mathbf{F}_1 = k \frac{Q\lambda d\beta a}{r'^3} (-a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k})$$

$$\mathbf{F}_1 = k \frac{Q\lambda a}{r'^3} \int_0^\pi d\beta (-a \cos \beta \mathbf{i} - a \sin \beta \mathbf{j} + r \mathbf{k})$$

$$\mathbf{F}_1 = k \frac{Q\lambda a}{r'^3} \left[\int_0^\pi -a \cos \beta d\beta \mathbf{i} + \int_0^\pi -a \sin \beta d\beta \mathbf{j} + \int_0^\pi r d\beta \mathbf{k} \right]$$

$$\mathbf{F}_1 = k \frac{Q\lambda a}{r'^3} \left[-a \sin \beta \Big|_0^\pi \mathbf{i} - a(-\cos \beta \Big|_0^\pi) \mathbf{j} + \beta \Big|_0^\pi r \mathbf{k} \right]$$

$$\mathbf{F}_1 = k \frac{Q\lambda a}{r'^3} [-2a \mathbf{j} + r\pi \mathbf{k}]$$

A força $d\mathbf{F}_2$ é:

$$d\mathbf{F}_2 = -k \frac{Qdq}{r'^3} \mathbf{r}_2$$

Escrevendo $\mathbf{r}_2$ em termos de $\mathbf{i}$, $\mathbf{j}$ e $\mathbf{k}$:

$$\mathbf{r}_2 = \mathbf{r} - \mathbf{a}_2$$

$$\mathbf{r}_2 = r\mathbf{k} - a(\cos \phi \mathbf{i} + \sin \phi \mathbf{j})$$

$$\mathbf{r}_2 = -a \cos \phi \mathbf{i} - a \sin \phi \mathbf{j} + r\mathbf{k}$$

A distribuição de carga na parte inferior seria:

$$\lambda = \frac{dq}{d\phi a} \Rightarrow dq = \lambda d\phi a$$

Ou seja,

$$d\mathbf{F}_2 = -k \frac{Q\lambda d\phi a}{r'^3} [-a \cos \phi \mathbf{i} - a \sin \phi \mathbf{j} + r\mathbf{k}]$$

$$\mathbf{F}_2 = -k \frac{Q\lambda a}{r'^3} \left[-a\mathbf{i} \int_{\pi}^{2\pi} \cos \phi d\phi - a\mathbf{j} \int_{\pi}^{2\pi} \sin \phi d\phi + r\mathbf{k} \int_{\pi}^{2\pi} d\phi \right]$$

$$\mathbf{F}_2 = -k \frac{Q\lambda a}{r'^3} [-a\mathbf{i} (\sin \phi \big|_{\pi}^{2\pi}) - a\mathbf{j} (-\cos \phi \big|_{\pi}^{2\pi}) + r\mathbf{k} (\phi \big|_{\pi}^{2\pi})]$$

$$\mathbf{F}_2 = -k \frac{Q\lambda a}{r'^3} [2a\mathbf{j} + r\pi\mathbf{k}]$$

A força resultante será então de:

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 = k \frac{Q\lambda a}{r'^3} [-2a\mathbf{j} + r\pi\mathbf{k}] - k \frac{Q\lambda a}{r'^3} [2a\mathbf{j} + r\pi\mathbf{k}]$$

$$\mathbf{F} = k \frac{Q\lambda a}{r'^3} [-2a\mathbf{j} + r\pi\mathbf{k} - 2a\mathbf{j} - r\pi\mathbf{k}]$$

$$\mathbf{F} = -k \frac{4Q\lambda a^2}{r'^3} \mathbf{j}$$

Por o sistema ser um cone temos que:

$$r'^2 = r^2 + a^2 \Rightarrow r' = \sqrt{r^2 + a^2} \Rightarrow r'^3 = (r^2 + a^2)^{3/2}$$

Além disso:

$$\lambda = \frac{q_t}{2\pi a}$$

Sendo q_t a carga total que pode ser escrita como a diferença das cargas superior e inferior: $q_1 - q_2$ Ou seja:

$$\mathbf{F} = -k \frac{4Q(q_1 - q_2)a^2}{2\pi a(r^2 + a^2)^{3/2}} \mathbf{j}$$

$$\mathbf{F} = -k \frac{2Q(q_1 - q_2)a}{\pi(r^2 + a^2)^{3/2}} \mathbf{j}$$

$$\mathbf{F} = -\frac{Q(q_1 - q_2)a}{2\epsilon_0\pi^2(r^2 + a^2)^{3/2}} \mathbf{j}$$

Ou seja:

$$\mathbf{E} = -\frac{(q_1 - q_2)a}{2\epsilon_0\pi^2(r^2 + a^2)^{3/2}} \mathbf{j}$$